

Abemaciclib With Endocrine Therapy in the Treatment of High-Risk Early Breast Cancer: ASCO Optimal Adjuvant Chemotherapy and Targeted Therapy Guideline Rapid Recommendation Update

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ASCO Rapid Recommendations Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the ASCO Guideline Methodology Manual. The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options.

BACKGROUND

In 2020, ASCO published a guideline on the selection of optimal adjuvant chemotherapy and targeted therapy for breast cancer.¹ Recently, Harbeck et al² reported results of updated analyses of data from monarchE, an open-label, randomized (1:1), two-cohort phase III trial that evaluated the addition of two years of abemaciclib, a cyclin-dependent kinase (CDK) 4/6 inhibitor, to physician's choice of endocrine therapy (ET, given for ≥ 5 years; $n = 2,808$) versus standard ET alone ($n = 2,829$) among patients with resected, high-risk, hormone receptor-positive, human epidermal growth factor receptor 2 (HER2)-negative, node-positive breast cancer, which led to the US Food and Drug Administration (FDA) approval of adjuvant abemaciclib. Significant improvements in the primary study end point of invasive disease-free survival (IDFS), now observed at two analysis timepoints for those receiving abemaciclib—median follow-up of 19 months and 27 months, respectively—constituted a strong signal for amending the 2020 guideline. This update addresses the use of abemaciclib in the adjuvant setting; data on other CDK 4/6 inhibitors in this setting are not addressed.

METHODS

A literature search was conducted to identify phase III trials on the use of abemaciclib in this patient population. No additional trials of adjuvant abemaciclib were identified. The Expert Panel reconvened to assess evidence from monarchE and to review and approve the amended guideline.

EVIDENCE REVIEW

Harbeck et al² reported that, in the primary outcome analysis (median follow-up of 19 months), the intent-

to-treat (ITT) analysis revealed that adjuvant abemaciclib plus ET significantly improved IDFS compared with ET alone (hazard ratio [HR], 0.71, 95% CI, 0.58 to 0.87; $P = .00090$). This benefit was maintained for IDFS in the additional follow-up analysis (median follow-up of 27 months; HR, 0.70, 95% CI, 0.59 to 0.82; $P < .0001$) with a 5.4% absolute improvement in 3-year IDFS. In analyses of Ki-67–defined subpopulations using a prespecified Ki-67 threshold of $\geq 20\%$, abemaciclib plus ET significantly improved IDFS among patients with high Ki-67 in both the ITT population and the cohort 1 population ($n = 2,003$; HR, 0.63; 95% CI, 0.49 to 0.80; $P = .0042$) that included patients with either ≥ 4 positive axillary lymph nodes (ALNs) or 1-3 positive ALNs and at least one of the following: tumor size ≥ 5 cm or histologic grade 3.

Of note, longer follow-up is needed to determine if the IDFS benefit of abemaciclib will be sustained over time. Overall survival (OS) data were reported in a subsequent letter to the editor.³ OS data remain immature but show no improvement (HR, 1.091; 95% CI, 0.818 to 1.455, in the ITT population and HR, 0.767; 95% CI, 0.511 to 1.152, in the FDA-indication population).

The incidence of $\geq$ grade 3 treatment-related adverse events (AEs) was higher in the abemaciclib plus ET arm (50%) versus in the ET-alone arm (16%); there was also a higher incidence of serious AEs in the abemaciclib (15%) versus the control arm (9%). The most frequent AEs of any grade in the abemaciclib plus ET arm were diarrhea (83.5%), fatigue (40.6%), and neutropenia (45.8%); in the control arm, the most frequent AEs of any grade were fatigue (17.8%), hot flush (23.0%), and arthralgia (37.9%). Venous

ASSOCIATED CONTENT

The companion to this article was published in the February 20, 2021 issue of *Journal of Clinical Oncology*. See accompanying article on page 685

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on November 17, 2021 and published at ascopubs.org/journal/jco on December 8, 2021; DOI <https://doi.org/10.1200/JCO.21.02677>

Evidence-Based Medicine Committee approval: November 15, 2021

thromboembolic events and interstitial lung disease each occurred in 3% of patients who received abemaciclib plus ET. Fatal AEs occurred in 0.8% of patients taking abemaciclib plus ET.

2021 UPDATED RECOMMENDATIONS

- On the basis of a secondary predefined analysis conducted by the FDA (https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/208716s006s007s008lbl.pdf), 2 years of abemaciclib (150 mg twice daily) plus ET may be offered to patients with hormone receptor–positive, HER2-negative, node-positive early breast cancer with a high risk of recurrence and a Ki-67 score of $\geq 20\%$ as determined by an FDA-approved test (Type: evidence-based, benefits outweigh harms; Evidence quality: moderate; Strength of recommendation: strong).
- The Panel also recommends, on the basis of analyses reported by Harbeck et al, that abemaciclib for 2 years plus ET for ≥ 5 years may be offered to the broader ITT population of patients with resected, hormone receptor–positive, HER2-negative, node-positive, early breast cancer at high risk of recurrence, defined as having ≥ 4 positive ALNs or as having 1–3 positive ALNs and one or more of the following features: histologic grade 3 disease, tumor size ≥ 5 cm, or Ki-67 index $\geq 20\%$ (Type: evidence-based, benefits outweigh harms; Evidence quality: moderate; Strength of recommendation: strong).

Qualifying Statements

Although exploratory analyses suggested similar HRs in favor of abemaciclib regardless of Ki-67 status, there were relatively few Ki-67 low tumors in monarchE. When discussing treatment options with patients, the potential benefits (improved IDFS) should be weighed against the potential harms (treatment toxicity and financial cost).

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GUIDELINE AND CONFLICTS OF INTEREST

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EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at www.asco.org/breast-cancer-guidelines.

EQUAL CONTRIBUTION

S.H.G. and R.A.F. were the Expert Panel coauthors.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/JCO.21.02677>.

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Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

The Expert Panel would like to thank Drs Lisa A. Carey and E. Claire Dees and the ASCO Evidence-Based Medicine Committee for their thoughtful reviews and insightful comments on this guideline. The following are the members of the ASCO Optimal Adjuvant Chemotherapy and Targeted Therapy for Early Breast Cancer Guideline Expert Panel: Mariana Chavez-MacGregor, MD, MSC; Amy H. Comander, MD; Zoneddy Dayao, MD; Andrea Eisen, MD; Rachel A. Freedman, MD, MPH; Ragisha Gopalakrishnan, MD; Stephanie L. Graff, MD; Michael J. Hassett, MD, MPH; Tari A. King, MD; Gary H. Lyman, MD, MPH; Raquel Nunes, MD; Cheryl L. Perkins, MD, RPh; Melinda L. Telli, MD; Maureen E. Trudeau, MD; Antonio C. Wolff, MD; and Sharon H. Giordano, MD, MPH. Finally, the Panel is grateful to Brittany E. Harvey for her assistance in completing the evidence summary for this rapid update.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Abemaciclib With Endocrine Therapy in the Treatment of High-Risk Early Breast Cancer: ASCO Optimal Adjuvant Chemotherapy and Targeted Therapy Guideline Rapid Recommendation Update**

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Employment: Firefly Health

Stock and Other Ownership Interests: Firefly Health

Research Funding: Puma Biotechnology (Inst), Eisai (Inst), Genentech/Roche (Inst)

No other potential conflicts of interest were reported.